KRISHNA ANNAVARAM

Generative AI Engineer

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PROFESSIONAL SUMMARY

Generative AI Engineer with 5+ years building production-grade LLM systems, RAG pipelines, and multi-agent orchestration frameworks across healthcare and enterprise domains with a track record of shipping systems that directly move clinical and operational metrics at scale. Delivered measurable outcomes including 91% retrieval accuracy on a hospital-deployed Graph-RAG Clinical Decision Support System, 45% reduction in unsafe clinical recommendations, 85% automation of patient intake workflows, and 50% reduction in physician review time through fine-tuned transformer models. Deep full-stack GenAI expertise spanning LangGraph, Azure OpenAI (GPT-4o), Neo4j Knowledge Graphs, FAISS and Pinecone vector search, LLM fine-tuning (LoRA, PEFT, RLHF), and zero-downtime MLOps pipelines on Azure, AWS, and GCP.

CORE TECHNICAL SKILLS

Generative AI & LLMs	Azure OpenAI (GPT-4o, GPT-4), LangChain, LangGraph, LangSmith, RAG Pipelines, Graph-RAG, Agentic AI, Multi-Agent Orchestration, A2A/MCP Protocols, Prompt Engineering, Chain-of-Thought, Few-Shot Learning
LLM Fine-Tuning	LoRA, QLoRA, PEFT, Adapter Tuning, Instruction Tuning, RLHF, Long-T5, BART, BERT, BioBERT, ClinicalBERT, LLaMA, Mistral, RoBERTa, ALBERT, Hugging Face Transformers
AI Agents & Pipelines	LangGraph Agent Workflows, Tool Use, Function Calling, ReAct Agents, Memory Management, Vector Search, Semantic Retrieval, Hybrid Dense-Sparse Retrieval, Sentence Transformers, FHIR-based EHR Integration
Cloud Platforms	Azure (Azure OpenAI, Azure ML, AKS, Cosmos DB, Azure Speech, Azure Functions, Azure DevOps, Databricks) · AWS (SageMaker, Lambda, S3, EC2, Bedrock, Glue, Athena, Kinesis, CodePipeline, CloudWatch) · GCP (Vertex AI, BigQuery, Dataflow, Dialogflow CX)
Vector DBs & Knowledge Graphs	FAISS, Chroma DB, Pinecone, Milvus, Weaviate, Neo4j (Knowledge Graphs), PostgreSQL pgvector, Redis Semantic Cache
MLOps & CI/CD	MLflow, Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps, Airflow, Kubeflow, Helm, Jenkins, CircleCI, Model Monitoring, A/B Testing, Canary Deployments
ML & Deep Learning	PyTorch, TensorFlow, scikit-learn, Keras, XGBoost, LightGBM, NLTK, TF-IDF, Cosine Similarity, CNN, RNN, LSTM, Transformers, GANs, NLP Pipelines, Named Entity Recognition, Text Classification
Programming & Data	Python (Expert), SQL, Bash, JavaScript, TypeScript, R · PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, IBM DB2 · Pandas, NumPy, Polars, SciPy, ETL Pipelines

PROFESSIONAL EXPERIENCE

Generative AI Engineer — May 2025 – Present

Ideate Technologies LLC | United States

- Led the end-to-end delivery of a Graph-RAG Clinical Decision Support System for U.S. hospital clients unifying Neo4j Knowledge Graphs, FAISS vector search, and Azure OpenAI GPT-4o into a real-time diagnostic reasoning pipeline that achieved 91% retrieval accuracy and reduced unsafe clinical recommendations by 45%, directly improving patient safety outcomes.
- Directed the full-stack development of a voice-based patient intake platform deployed across hospital workflows integrating Azure OpenAI, Azure Speech-to-Text, LangGraph multi-agent orchestration, and Cosmos DB to automate 85% of clinical data capture and reduce patient intake cycle time by 60%.
- Secured enterprise-grade compliance across hospital-facing AI systems by architecting LangGraph A2A/MCP multi-agent workflows with live FHIR-based EHR integration enabling stateful, context-aware clinical interactions at scale while meeting U.S. healthcare regulatory standards.

- Drove a 50% reduction in physician review time by leading the fine-tuning of Long-T5 and BART transformer models with PEFT and LoRA on real clinical corpora delivering structured nurse-to-doctor handoff summaries with 92% completeness across active hospital use cases.
- Established a zero-downtime GenAI MLOps infrastructure for hospital-grade AI services using AKS, Azure ML, Azure OpenAI, MLflow, Terraform, and GitHub Actions cutting deployment effort by 70%, accelerating release cycles by 40%, and sustaining 99.9% uptime across all production clinical systems.

Graduate Assistant, Generative AI August 2024 – May 2025

University of North Texas | Denton, TX

- Built and deployed an AI-powered research assistant into UNT's graduate program infrastructure combining FAISS vector search, PostgreSQL, and Redis into a scalable multi-turn retrieval system that delivered 88–92% retrieval relevance across thousands of student and faculty queries, replacing manual literature search.
- Developed a full-stack RAG pipeline on Azure Cloud using LangGraph, Sentence Transformer embeddings, and hybrid dense-sparse retrieval enabling accurate semantic search across 10,000+ UNT academic documents with sub-second response times under concurrent multi-user load.
- Designed and deployed a context-aware LLM system with dynamic prompt engineering and real-time context injection reducing repeated query resolution time by 40–45% and decreasing manual faculty intervention by streamlining student self-service access to research resources.
- Improved vector retrieval performance by 35% by systematically tuning FAISS index parameters and implementing a Redis caching layer maintaining consistent low-latency responses across concurrent sessions during peak academic demand periods.

AI Engineer November 2022 – December 2023

Sun Pharma (via CreativeSense Pvt Ltd) | India

- Transformed Sun Pharma's unstructured drug report backlog into a queryable structured asset deploying a production BioBERT named entity recognition pipeline on GCP that ingested thousands of daily pharma documents, extracted drug names, dosages, adverse events, and indications at scale, and wrote clean records to BigQuery by eliminating 60% of manual data entry and unblocking the analytics division.
- Rebuilt the end-to-end text preprocessing stack with Hugging Face Transformers, improving clinical NLP model accuracy by 38% integrating medical tokenization, negation detection, and domain normalization to ensure BioBERT received clean, consistent signal across thousands of heterogeneous pharma documents at production scale.
- Scaled BioBERT and ClinicalBERT model delivery by 45% by migrating training and batch inference to GCP Vertex AI auto-scaled endpoints replacing fragile manual scripts with versioned, reproducible ML pipelines that powered Sun Pharma's full medico-marketing analytics stack reliably across 365-day operations.
- Reduced end-to-end pharma analytics reporting latency by 50% by re-architecting the data delivery layer with GCP Dataflow streaming ETL routing named entity recognition-extracted drug data into BigQuery in real time and enabling Sun Pharma's analytics team to query clean, structured records within minutes of raw document arrival.

Machine Learning Consultant August 2021 – November 2022

Cognizant Technology Solutions | India

- Deployed 3 automated batch pipelines on AWS processing daily incentive compensation files for a U.S. medical device sales organization using Python, S3, Lambda, and SageMaker Batch Transform cutting commission calculation cycle time by 40% and guaranteeing county-level payout calculations within every 24-hour window.
- Eliminated 55% of manual business rules processing by engineering an NLTK document parsing pipeline that automatically extracted compensation rules, territory hierarchies, and manager reassignments from daily flat-file feeds removing error-prone human interpretation across 365-day continuous operations.
- Increased incentive payout forecasting accuracy by 35% by deploying XGBoost classification models on AWS SageMaker trained on multi-year sales performance data to predict commission attainment tiers across U.S. county-level territories, retiring legacy spreadsheet-based calculations entirely.
- Led CI/CD pipeline ownership and junior engineer mentorship across a cross-functional ML team standardizing AWS CodePipeline, Git branching, and CloudWatch alerting to achieve zero-downtime deployments across all 3 production incentive pipelines running 24/7/365.

Machine Learning Engineer May 2020 – August 2021

Lemoius Pvt. Ltd. | India

- Pioneered a real-time content-based job recommendation engine for a 20,000+ profile marketplace by implementing TF-IDF vectorization and cosine similarity ranking over candidate profiles and job description text, lifting candidate-to-role match relevance by 35% and replacing manual recruiter shortlisting.
- Reduced candidate profile processing time by 30% by engineering an end-to-end NLP preprocessing pipeline with NLTK executing tokenization, stopword removal, lemmatization, and n-gram extraction to transform raw candidate data into clean, model-ready feature vectors at scale.
- Accelerated recruiter shortlisting efficiency by 25% by productionizing a real-time top-N candidate ranking system using scikit-learn TF-IDF and cosine similarity scoring, delivering ranked candidate lists for every new job posting with zero manual intervention.
- Reduced recommendation model iteration cycles by 40% by standardizing reproducible ML experiments with scikit-learn pipelines and Git version control, enabling the team to evaluate, validate, and ship updated matching models within a single sprint.

EDUCATION

Master of Science in Data Science December 2025

University of North Texas | Denton, TX | Concentration: Applied NLP & Generative AI

Bachelor of Technology in Computer Science & Engineering May 2021

Kalasalingam University | India | Major: Machine Learning & Deep Learning